REAL ESTATE TRACK · AI HARVEST: TURNING EXPERT KNOW-HOW INTO REUSABLE INTELLIGENCE

instill.

Expert judgment, instilled into every new case.

Our slice of the challenge: one end-to-end loop that captures an expert's judgment, gets it approved, and reuses it on the next similar case. Built and running in Singapore customs brokerage.

THE PART OF THE PROBLEM WE SOLVE

One complete learning loop, not a whole platform

1 · Capture

Expert decisions on real cases, plus existing documents

2 · Approve

Nothing is used until a manager approves it

3 · Feed back

Approved records join the agents' knowledge

4 · Reuse

They return as precedents on the next similar case

All four steps run end to end today, covered by automated tests. Everything else in this deck builds on this loop.

1 · THE BUSINESS PROBLEM

Experts have the SOP. Juniors still miss the check.

Recall fails, not knowledge

Guides and pop-ups exist. People still miss the check at the moment of work.

Corrections disappear

Seniors fix the same mistake again and again. The reason stays on a marked-up printout.

People must stay in charge

Seniors won't drop human review, or adopt a tool that adds to their work.

REAL INCIDENT

A car remote control needed a telecom permit. The number was right; the permit had expired. Customs accepted the filing, the goods were held, and the shipment was lost. Even the senior reviewer missed it.

Source: interview with a customs brokerage supervisor who trains new staff and reviews every declaration (translated).

2 · WHOSE EXPERTISE, AND HOW WE CAPTURE IT

Captured from approvals and documents

UNDOCUMENTED

Senior judgment

Corrections that today live on a marked-up printout

DOCUMENTED

Existing documents

SOPs, handover notes and emails: PDF, DOCX, EML, CSV

AUTOMATIC

Captured at approval

Each decision is saved with its case; a rejection must say why

GATED

Imported as pending

Text extracted and tagged to a skill; unused until approved

REUSABLE

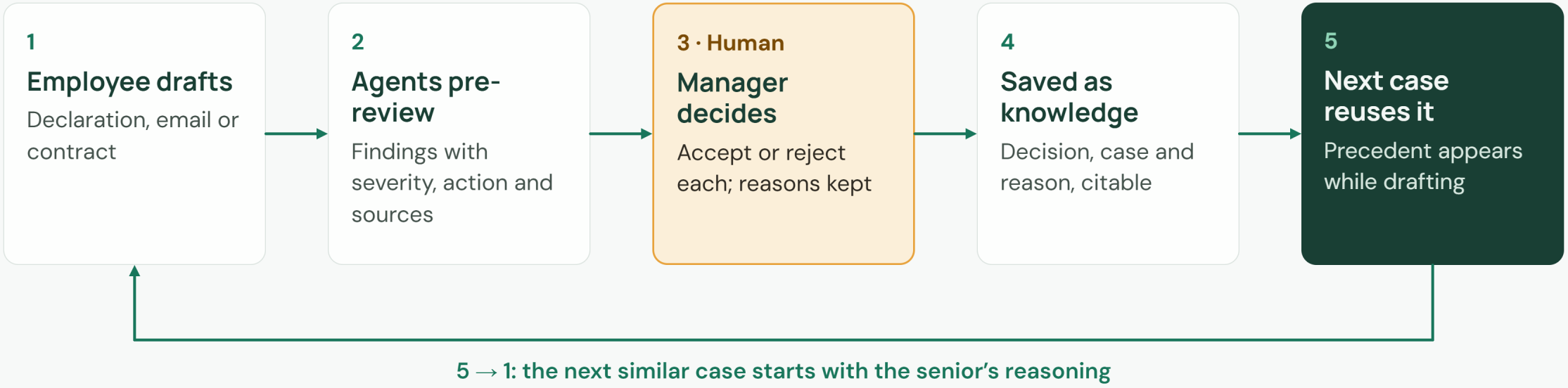
Approved knowledge, by skill

- Electronics & telecoms
- Food & produce
- Automotive
- General cargo

Every record keeps its source, author, case and approval status.

3 · HOW EXPERTISE BECOMES REUSABLE

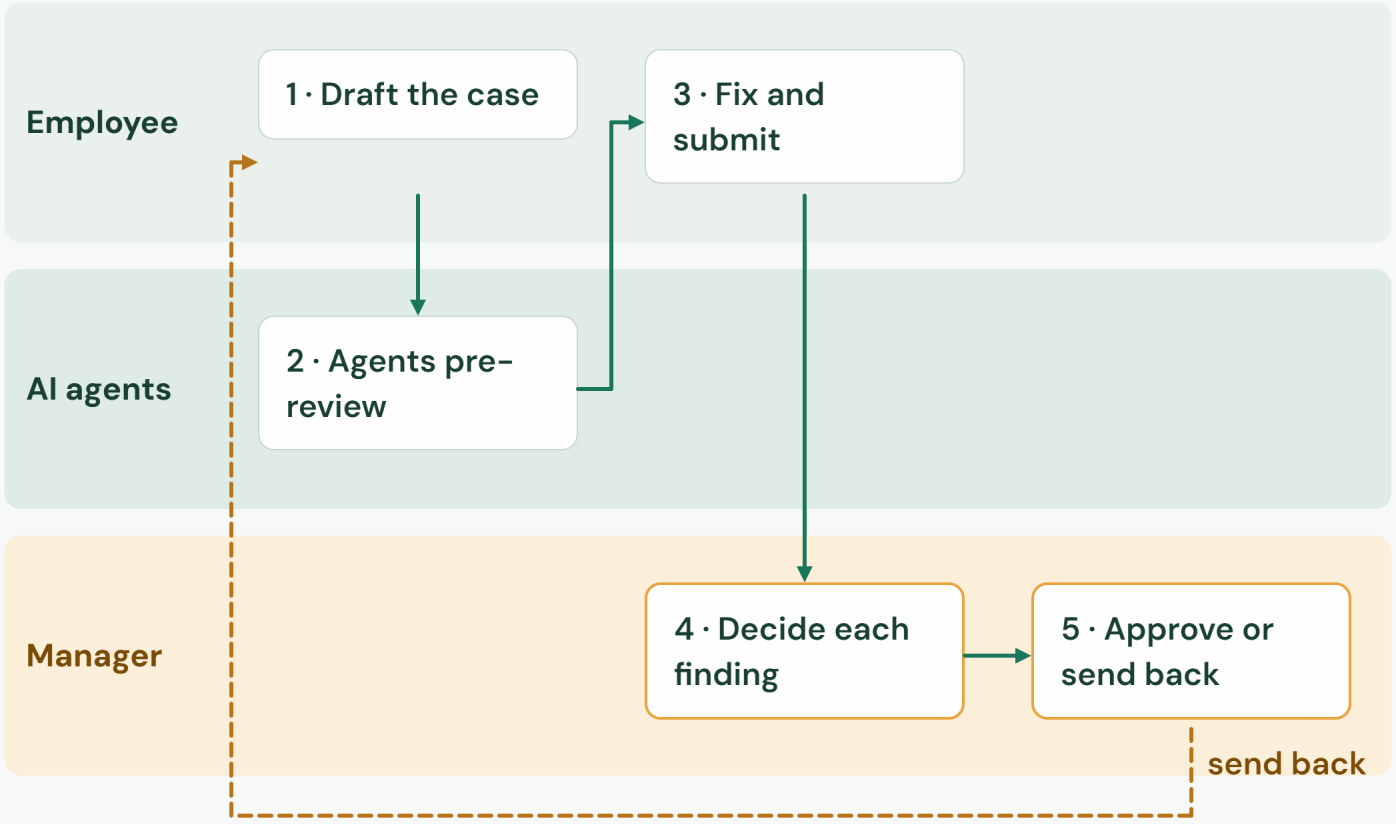
One decision, reused on every similar case



Seniors review exactly as they do today. The only new step is one sentence of reasoning, and it pays back on every similar case.

4 · HOW THE MANAGER SUPERVISES

AI checks first. A manager always decides.



Always a human

- Every finding needs a decision; a rejection, a reason
- Only a manager approves, never the author
- Editing a case clears its review

The AI never

- Approves, submits or sends anything
- Cites a source nobody approved

For Keppel, each pill would also set its own escalation triggers, such as a critical finding going to a senior engineer.

KEEPING KNOWLEDGE FREE OF HALLUCINATION

Six guardrails, all in the running build

Approved sources only

Pending imports are never retrieved or cited.

Citations checked

Model output keeps only citations that match a real source.

Rules are code

Date and permit checks are computed, not generated.

Input is untrusted

Prompts treat documents as data, never as instructions.

A human decides

Every finding needs a manager’s decision before approval.

Visible fallback

If the model fails or times out, the local advisor answers and says so.

These make every piece of advice traceable to an approved source. They don’t prove it correct, which is why a person still decides.

5 · WHY IT IS MORE THAN A CHATBOT

Built into the work, not waiting to be asked

	Chatbot or search	Instill
When it helps	When someone thinks to ask	Before approval, on every case
What it knows	Static documents	Documents plus every reviewed decision
Who decides	The user, alone	A manager, per finding, with a reason
Where advice comes from	Often unclear	Approved sources, cited on every finding
How it learns	Only when a document is edited	Each decision becomes a governed record
Across domains	One general answer	Several specialist skills on one case

6 · HOW NEW EXPERTISE IS GOVERNED

Nothing reaches the agents until a manager approves it

1 · Captured
An import or decision enters as a candidate

→

2 · Validated
A manager approves it, with a reason

→

3 · Released
Agents may now retrieve and cite it

→

4 · Re-checked
Every later use is reviewed by a manager

Built today

- Pending → approved status on every record
- Decisions keep reviewer, reason and case
- Activity history of who decided what
- Workspace export: the knowledge stays yours
- Optional ADP sync, with publish status

Next, for production

- Versioned pill releases with rollback
- An owner and scope rules for each pill
- Role-based access and single sign-on
- Retention rules and tamper-evident audit

7 · FROM CUSTOMS TO REAL ESTATE, AND ACROSS THE ENTERPRISE

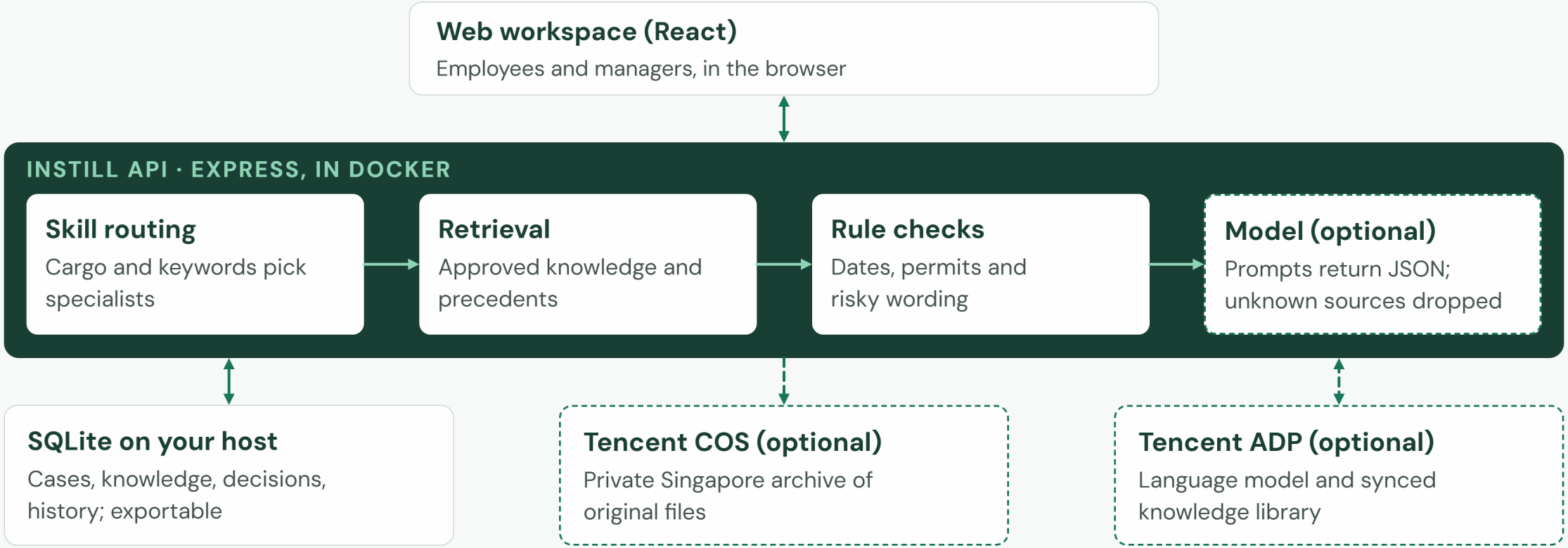
Same engine, real estate examples

Real estate situation	How Instill handles it
A veteran reads a building and sees where to add lettable area	Their reasoning is captured as decisions on real cases, then reused on similar ones
An energy manager's answers to management are never written down	Reasoning given at sign-off is saved with the case and cited next time
Is an anomaly a one-off, or a recurring problem?	Precedents show when an issue has come up before; automatic detection is next
An issue that spans energy and operations	Several skills on one case; works today for a car part with a radio inside
Keep a human in the loop; no hallucinated knowledge	A manager decides every finding; only approved, cited sources are used

One team → **Skills shared across assets** → **Enterprise, across countries**

8 · IMPLEMENTATION: ARCHITECTURE

Built on current technology, owned by you



Dashed = optional Tencent Cloud services. Without them, Instill runs fully on the local advisor, and it takes over if a model call fails or times out.

8 · IMPLEMENTATION: ROADMAP

Working today, ready for a supervised pilot

NOW · BUILT

Working engine

- Review loop, knowledge base and agent chat
- Docker images and Tencent ADP adapter
- 51 automated tests passing

PILOT · 0–3 MONTHS

One team, one skill

- For example, energy anomaly reviews
- Seed it with SOPs and expert decisions
- Measure review time and repeat issues

SCALE · 3–12 MONTHS

Learning across assets

- Flag recurring versus one-off issues
- Suggest new skills from novel decisions
- Versioned releases, sign-on, more assets

Pilot and scale timings are our proposal. No production impact has been measured yet; the pilot is where we would measure it.

Expert judgment, instilled into every new case.

Seniors decide as they do today. Every decision becomes cited, governed guidance for the next person.

INSTILL · AI HARVEST